

Lecture 34

4/17/26
JRA

HW - due today (or by end of weekend)

Finish nuclear imaging, then start MRI

Looking at two techniques for 3D imaging

SPECT - $Tc-99m$

Single γ ray

camera(s) rotate around patient
needs collimator

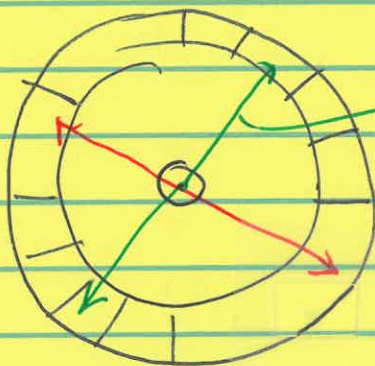
PET - ^{18}F

Position $\rightarrow$ 2 γ rays

PET Detector ring

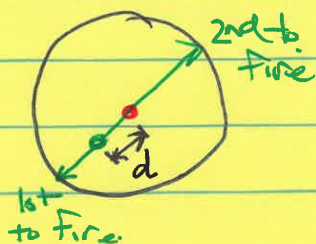
Line of response

/ stationary
No collimator



Absence of collimator is an improvement over SPECT
Another improvement is TOF PET

TOF



Here, hot spot closer to one side of PET ring than other side
 $\Rightarrow$ One detector, ^{closer} will fire first, one ^{further} will fire second

Calculate distance, ^d of hot spot from center of ring

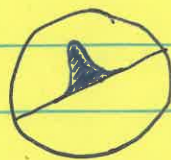
Punch Line $d = \frac{c \Delta t}{2}$

$\Delta t = t_1 - t_2$ difference in firing time

In theory, TOF PET gives you the location of decay event/hot spot

This is the goal of reconstruction, so TOF PET theoretically supplants reconstruction.

Current electronics NOT quite good enough to determine "d" precisely but it gives enhanced probability



Derivation

$$x_2 = ct_2$$

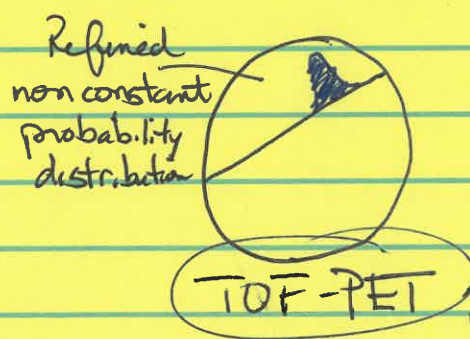
$$x_1 = ct_1$$

$$\Delta x = x_1 - x_2 = ct_1 - ct_2 = c(t_1 - t_2) = c\Delta t$$

Center is at avg

$$d = \frac{x_1 + x_2}{2} - x_2 = \frac{x_1 + x_2 - 2x_2}{2} = \frac{x_1 - x_2}{2} = \frac{c\Delta t}{2}$$

Current timing resolution of detectors just refines the event location



This Improves the signal-to-noise markedly

Very common now

Most PET instrument use TOF-CT

State-of-Art = TOF PET / CT

↑
high quality functional

anatomical

How do you reconstruct images?

Reconstruction of 2D/3D images in nuclear medicine

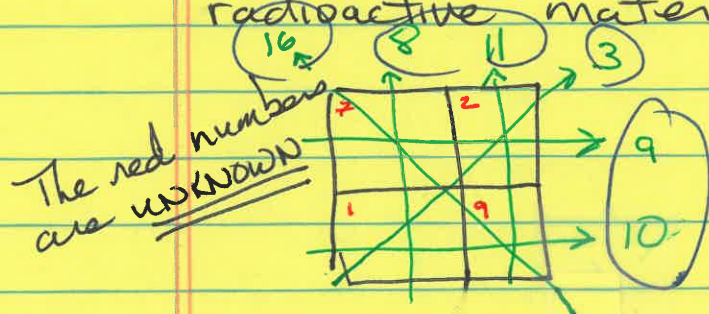
FBP - CT

Iterative reconstruction is gold std in nuclear medicine

Use iterative approach because take fewer images

ACTIVITY 12

Goal is to get the distribution of activity radioactive material in body



Collect 6 projections

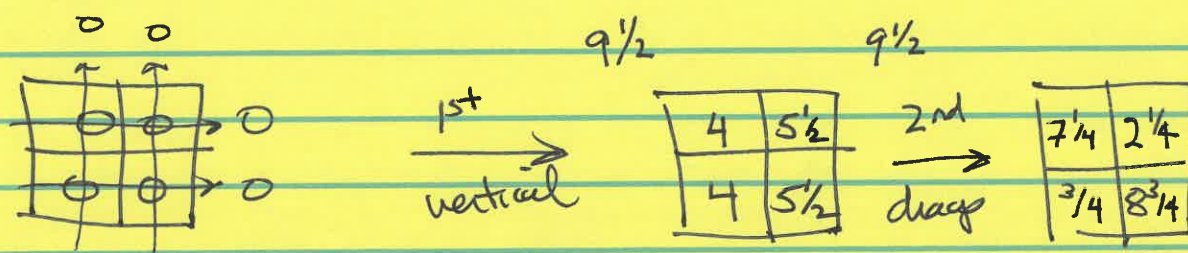
Green numbers are the data

= known

Know projection results

Can you iterate to get object intensity?

Start w/ a guess: every picture is zero in this scenario.



Compute projections from guess, then correct based on actual data, to improve guess

Compare 0 to 8

too low, so bump up numbers up uniformly by spreading real data over both pixels

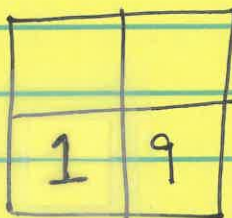
Compare 0 to 11

add back 5.5

Diagonals

Remove 6.5 w/ even split

3rd
→
horiz



Right
answer

Real life: noisy, many projections but still get good answer

Projected: Too low → raise
Too high → lower

Looked at flow charts

Classic apps: brain activity

Use FDG labeled w/ ^{18}F

Uptaken by metabolically active areas

Looked at brain scans $\left\{ \begin{array}{l} \text{Dementia dark} \\ \text{Cancer bright} \end{array} \right.$

Bone screening methyle disphosphanate

VIDEOS

^{18}F

Dr. Klose
Lung

Summary

Emission analog

Functional complement

Planar & tomography

MRI

MRI is an unusual technique

(1) It uses radio frequency $^{(RF)}$ radiation

RF = low energy $\Rightarrow$ safe (doesn't ionize atoms)

(2) Functional & anatomic info

Ex: use fMRI to look at what med does to ^{brain} activity

(3) Get superb soft-tissue contrast
There are many contrast parameters
(to be discussed later)

US - strength of echo

Nuc - amount of activity

(4) Resolution is excellent

MRI does have weaknesses

(1) Slow

compare to US which
is real-time

(2) Expensive

(3) Metal implants not allowed

Look at protons, typically hydrogen nuclei

Uses magnetism of proton

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